Problem G. Gungnir Fracture

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Geopelia is collecting the fragments of *Gungnir*. *Gungnir* has 2^n types of fragments, numbered $0, 1, \cdots, 2^n - 1$. To restore it, for every $0 \le i < 2^n$, *Geopelia* needs exactly a_i fragments marked i.

To obtain these fragments, *Geopelia* may perform the following operation any number of times:

- Choose two integers m and k satisfying $0 \le m < 2^n$ and $0 \le k < 2^n$.
- Obtain one fragment marked with each of $0 \oplus k, 1 \oplus k, \cdots, m \oplus k$.

Recall that $\oplus$ denotes the bitwise exclusive OR of two non-negative integers.

Help *Geopelia* find the minimum number of operations required to restore *Gungnir*. Note that *Geopelia* **cannot** remove any fragment she has obtained.

Input

The first line of the input contains an integer T ($1 \le T \le 10^4$) indicating the number of test cases. For each test case:

- The first line contains an integer n ($0 \le n \le 19$).
- The second line contains 2^n integers $a_0, a_1, \ldots, a_{2^n-1}$ ($0 \le a_i \le 10^6$), where a_i is the required number of fragments marked i.

It is guaranteed that the sum of 2^n over all test cases does not exceed 2^{19}.

Output

For each test case, output one integer representing the minimum number of operations.

Example

standard input	standard output
5 0 100000 1 114 514 2 1 1 1 1 3 0 13 0 0 0 7 2 1 4 1 2 2 1 3 3 3 3 1 0 0 0 0 0 0 0	100000 514 1 21 3